ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** a force acts on a body and the body is displaced in the direction of the force	W = F x s.
2	**(c)** kinetic energy	W = change in KE.
3	**(d)** makes work easier by changing the magnitude or direction of force	Machines help us.
4	**(d)** joule	1 J = 1 N m.
5	**(a)** can neither be created nor destroyed but only changed from one form to another	Law.
6	**(b)** load / effort	MA.
7	**(c)** zero	Force is perpendicular to displacement.
8	**(c)** rate of doing work	P = W/t.
9	**(d)** fulcrum	Pivot.
10	**(c)** 50 J	F s.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Product of force and displacement in the direction of force (1); Joule (1).	1+1
12	Work done on a body equals the change in its kinetic energy (2).	2
13	Changes the direction of the force (1); Convenient to pull downward (1).	1+1
14	When force is opposite to displacement (1); Friction on a sliding block (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lifting: force and displacement same direction, positive (1); Lowering: force upward, displacement downward, negative (1); W = mgh = 60 J (1).	1+1+1
16	Energy = 0.1 kW x 10 h = 1 kWh (1); 1 unit (1); Cost = 6 rupees (1).	1+1+1
17	Height = 2.4 m (1); Work = 960 J (1); P = 120 W (1).	1+1+1
	OR: Work to accelerate from rest: W = F s (1); F = ma, v squared = 2as (1); KE = (1/2) m v squared (1).	1+1+1
18	l = sqrt(30 squared + 40 squared) = 50 cm (1); MA = l/h (1); = 50/30 = 1.67 (1).	1+1+1
	OR: Work by force = 60 J (1); Work by friction = -24 J (1); Net work = 36 J (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	W = F s cos(theta) (1); Along: positive (1); Opposite: negative (1); Perpendicular: zero (1); Example 60 J (1).	1+1+1+1+1
	OR: At the extreme: PE max, KE zero (1); At the mean: KE max, PE zero (1); Total constant (1); Air resistance reduces amplitude (1); Energy converted to heat (1).	1+1+1+1+1
20	W = F s (1); v squared - u squared = 2as (1); W = (1/2)m v squared - (1/2) m u squared (1); 100 = (1/2)(2) v squared (1); v = 10 m/s (1).	1+1+1+1+1
	OR: Work per min = 1000 x 10 x 30 = 300000 J (1); P = 5000 W (1); Input = 5000/0.75 (1); = 6667 W (1); Efficiency = output/input (1).	1+1+1+1+1

Mistakes to avoid:

- Saying work is done by holding a load or by pushing a wall; without displacement, work is zero.
- Treating work done by friction or by a force perpendicular to the motion (a coolie walking with a load on his head) as positive.
- Writing $KE = mv^2$ or taking $KE \propto v$; doubling the speed makes the kinetic energy four times.
- Not converting units: minutes to seconds in $P = W/t$, kW to W, or 20 cm of stairs to metres.
- Believing that a machine (lever, ramp) saves work; it only reduces the effort by increasing the distance moved by the effort.

- Using $1\ \text{kWh} = 1000$ J instead of 3.6×10^6 J, or calling kWh a unit of power.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** watt	1 W = 1 J/s.
2	**(c)** between the load and the effort	e.g. scissors, seesaw.
3	**(d)** opposite to the displacement	e.g. friction.
4	**(b)** 3.6 x 10^6 J	1000 W x 3600 s.
5	**(b)** second class	Load between fulcrum and effort.
6	**(c)** kinetic energy	Motion.
7	**(c)** 50 W	W/t.
8	**(d)** third class	Effort between fulcrum and load.
9	**(d)** potential energy	Stored energy.
10	**(d)** 9 J	0.5 x 2 x 9.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Energy can neither be created nor destroyed (1); It can only be converted from one form to another (1).	1+1
12	Scissors: first class (1); Wheelbarrow: second class; tweezers: third class (1).	1+1
13	KE = (1/2) m v squared (1); = 10 J (1).	1+1
14	No displacement (1); W = F x s with s = 0 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Work done against gravity: W = F s (1); F = mg, s = h (1); PE = mgh (1).	1+1+1
16	Pivoted bar (1); Load arm and effort arm are distances from the fulcrum (1); MA = effort arm/load arm (1).	1+1+1
17	(a) Mechanical to electrical (1); (b) Electrical to sound (1); (c) Chemical to heat and light (1).	1+1+1
	OR: PE lost = KE gained (1); mgh = (1/2) m v squared (1); v = sqrt(2 x 10 x 2.5) = 7.07 m/s (1).	1+1+1
18	Effort x effort arm = load x load arm (1); For equilibrium (1); Longer effort arm gives MA > 1 (1).	1+1+1
	OR: KE = (1/2) x 1000 x 400 (1); = 200000 J (1); Work done by brakes = -200000 J (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Start: KE 800 J, PE 0 (1); At 10 m: PE 400 J, KE 400 J (1); Top at 20 m: PE 800 J, KE 0 (1); Total energy 800 J throughout (1); Conservation of mechanical energy (1).	1+1+1+1+1
	OR: Hydro: PE to KE to electrical (1); Car: chemical to heat to mechanical (1); Solar cell: light to electrical (1); Forms of energy listed (1); Total energy conserved (1).	1+1+1+1+1
20	First class: fulcrum in the middle (seesaw, scissors) (1); Second class: load in the middle (wheelbarrow, nutcracker) (1); Third class: effort in the middle (tweezers, fishing rod) (1); MA > 1, > 1 and < 1 (1); Positions of fulcrum, load, effort (1).	1+1+1+1+1
	OR: Work is energy transferred (1); Power is rate (1); J and W (1); P = 3600/120 (1); = 30 W (1).	1+1+1+1+1

Mistakes to avoid:

- Saying work is done by holding a load or by pushing a wall; without displacement, work is zero.
- Treating work done by friction or by a force perpendicular to the motion (a coolie walking with a load on his head) as positive.
- Writing $KE = mv^2$ or taking $KE \propto v$; doubling the speed makes the kinetic energy four times.
- Not converting units: minutes to seconds in $P = W/t$, kW to W, or 20 cm of stairs to metres.
- Believing that a machine (lever, ramp) saves work; it only reduces the effort by increasing the distance moved by the effort.
- Using $1\ \text{kWh} = 1000$ J instead of 3.6×10^6 J, or calling kWh a unit of power.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 1	Only changes direction.
2	**(c)** (1/2) m v squared	KE.
3	**(c)** 200 J	mgh.
4	**(d)** 2	Ideal.
5	**(d)** m g h	PE.
6	**(c)** decreases, increases	Energy converts.
7	**(b)** reducing the effort needed	Longer distance, smaller force.
8	**(d)** elastic potential energy	Elastic PE.
9	**(b)** constant	PE + KE conserved.
10	**(a)** l / h	Ideal MA.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ratio of load to effort (1); No unit (1).	1+1
12	Rate of doing work (1); Watt (1).	1+1
13	Kilowatt hour (kWh) (1); 1 kWh = 3.6 x 10^6 J (1).	1+1
14	PE = mgh (1); = 500 J (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	KE = PE at top (1); (1/2)(0.5)(100) = 0.5 x 10 x h (1); h = 5 m (1).	1+1+1
16	KE: moving car, flowing water (1); PE: stretched spring, water in a dam (1); KE motion; PE position or state (1).	1+1+1
17	Work for 20th floor is twice (1); Power for 10th floor: W/t (1); Power for 20th: 2W/2t = W/t, same (1).	1+1+1
	OR: Work = mgh = 100000 J (1); P = W/t (1); = 1000 W (1).	1+1+1
18	Some energy is always lost as heat due to friction (1); A machine cannot give more output than input (1); Law of conservation of energy (1).	1+1+1
	OR: Efficiency = useful work/total work x 100 (1); Friction lowers it (1); Always less than 100 per cent (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fixed: MA 1, changes direction (1); Movable: MA 2 (1); Combination increases MA (1); Used to lift heavy loads (1); Rope friction reduces efficiency (1).	1+1+1+1+1
	OR: PE at top 200 J (1); v = 20 m/s (1); KE at ground 200 J (1); At 5 m: PE 50 J, KE 150 J (1); Total 200 J (1).	1+1+1+1+1
20	Work in = work out (1); F x 6 = 1000 x 1.5 (1); F = 250 N (1); MA = 4 (1); Longer ramp, smaller force (1).	1+1+1+1+1
	OR: Work out cannot exceed work in (1); Effort x 1 = 100 x 0.2 (1); Effort = 20 N (1); MA = 5 (1); Smaller force but larger distance (1).	1+1+1+1+1

Mistakes to avoid:

- Saying work is done by holding a load or by pushing a wall; without displacement, work is zero.
- Treating work done by friction or by a force perpendicular to the motion (a coolie walking with a load on his head) as positive.
- Writing $KE = mv^2$ or taking $KE \propto v$; doubling the speed makes the kinetic energy four times.
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